

## Highlights

### **A Reproducible Prostate Whole-Slide Histopathology Workflow: A DIAGSET Evaluation**

Bruno Henrique Nunes, Ana Estela Antunes da Silva

- End-to-end DIAGSET workflow with auditable intermediate artifacts.
- DIAGSET evaluation with patient-level splits and train-derived stain state.
- Metric-specific normalization findings across Dice, AUROC, and patch AvAcc.
- Negative-patient false positives remain the main limitation.

# A Reproducible Prostate Whole-Slide Histopathology Workflow: A DIAGSET Evaluation

Bruno Henrique Nunes<sup>a,\*</sup>, Ana Estela Antunes da Silva<sup>a</sup>

<sup>a</sup>*School of Technology, University of Campinas (UNICAMP), Rua Paschoal Marmo, 1888 - Jd. Nova Italia, Limeira, 13484-332, Sao Paulo, Brazil*

## ARTICLE INFO

### Keywords:

digital pathology  
whole-slide imaging  
prostate cancer  
reproducibility  
stain normalization  
ensemble inference

## ABSTRACT

Whole-slide histopathology experiments can fail before model fitting through annotation ambiguity, contaminated patches, patient leakage, color variability, incompatible artifacts, and validation choices that leak into the test set. This work presents a reproducibility-oriented workflow for prostate cancer segmentation in digital pathology and evaluates it on the DIAGSET prostate histopathology dataset. The workflow converts slide and annotation pairs into patch datasets, performs manual-review-guided graph cleaning, creates patient-disjoint training, validation, and test splits, applies stain normalization from training-derived state, reduces only the training split by embedding-based smart sampling, trains a fixed registry of segmentation models, optimizes a two-stream ensemble on validation patients, and evaluates the frozen recipe on held-out test patients. In the DIAGSET experiment, 419 valid image–annotation cases yielded 1,186,778 retained patches across 393 patients; the test set contained 79 patients and 387,325 patches. The analysis is descriptive rather than a statistical superiority test among normalization methods. The non-normalized setting produced the largest point estimates for pixel-level micro Dice (0.8415; 95% confidence interval, 0.7767–0.8987) and patient-level macro Dice (0.7540; 0.6710–0.8356). Ruifrok produced the largest area under the receiver operating characteristic curve point estimate (0.9579) and positive-patient Dice (0.7985; 0.7189–0.8688), and Macenko produced the largest patch balanced accuracy before patient suppression (0.9009). However, all settings produced false positives in at least 11 of 41 negative test patients. The results support an auditable single-dataset experimental protocol, while showing that clinical interpretation remains limited by false-positive control, validation-set optimization burden, and the absence of external validation.

## 1. Introduction

Deep-learning pipelines for whole-slide histopathology are shaped as much by data preparation as by the final neural architecture. A model can appear effective while relying on ambiguous annotations, repeated patient information across splits, contaminated training patches, preprocessing state derived from evaluation data, or test-time thresholds adjusted after inspecting held-out performance. These risks are especially consequential in computational pathology, where gigapixel images are converted into many smaller patches and where patient-level correlation can make naive patch-level evaluation overly optimistic [24, 5].

This paper reports a workflow designed to make those experimental decisions explicit and auditable. Rather than treating cancer detection as a single training script, the repository separates slide ingestion, patch extraction, review-oriented cleaning, patient-level split generation, stain-normalization state construction, sanity checking, training-sample selection, model fitting, validation-only ensemble optimization, and frozen test inference. The resulting artifact chain is intended to reduce silent reuse of stale data, preserve image–mask alignment, and make final results traceable to specific configurations and intermediate decisions [32, 20].

The empirical evaluation uses DIAGSET, a public prostate histopathology dataset with image and annotation material suitable for cancer versus non-cancer analysis [14]. The main experimental question is not whether one color-normalization method is universally superior, but whether a complete, provenance-preserving workflow can compare normalization strategies under a fixed training, validation, and test protocol. Five runtime settings are evaluated: no color normalization, Reinhard, Ruifrok, Macenko, and Vahadane [23, 25, 15, 30].

The descriptive results show that the largest point estimate depends on the evaluation lens. Pixel- and patient-level Dice were largest without normalization, AUROC and positive-patient Dice were largest with Ruifrok, and DIAGSET-style patch AvAcc was largest with Macenko. This disagreement is not treated as a nuisance result or as evidence of statistical superiority by one method; it is the central interpretation. In segmentation tasks with both positive and negative patients, micro Dice, macro Dice, AUROC, patch-recognition metrics, and negative-patient false-positive control answer different scientific questions [26, 9, 27]. The paper therefore emphasizes metric-specific interpretation and conservative scope rather than a single winning configuration.

## 2. Related Work

Whole-slide analysis often starts with quality control, tissue detection, and patch extraction because WSIs are too large to process directly and can contain acquisition artifacts,

\*Corresponding author

 bruno.nunes.1987@gmail.com (B.H. Nunes); aeasilva@unicamp.br (A.E.A.d. Silva)  
ORCID(s):

background regions, annotation inconsistencies, and tissue fragments that degrade downstream learning [31, 18, 13]. The present workflow follows this tradition but makes patch identities, HDF5 row alignment, and provenance checks explicit, so that downstream learning consumes a stable dataset contract rather than an informal folder of image files [11, 32]. This emphasis is important in DIAGSET because the available prostate histopathology material supports local cancer versus non-cancer analysis, but results from one public dataset cannot establish general validity across organs, scanners, annotation protocols, or cancer types [14].

Color variation is another persistent issue in histopathology. Stain normalization and stain augmentation have been studied as ways to reduce scanner- and protocol-dependent appearance shifts, but their effect can be dataset- and metric-dependent [29]. The workflow therefore stores training-derived normalization state and applies the selected normalization at runtime, allowing Reinhard, Ruifrok, Macenko, Vahadane, and non-normalized inference to share the same patient split [23, 25, 15, 30]. For Vahadane, this paper reports the configured fixed-source approximation, not exact per-patch source fitting.

Semantic segmentation models and ensembles provide the predictive core of many computational-pathology systems, but their reported performance depends on the surrounding experimental contract as much as on the architecture itself [13, 6, 4]. In this work, a fixed registry of segmentation architectures is trained under each normalization setting, and the final prediction stage combines selected models through a validation-optimized two-stream recipe. Test-time augmentation is used inside this frozen inference procedure, but no test patient is used to choose model weights, thresholds, or post-processing parameters [17].

Evaluation in segmentation must also distinguish the question answered by each metric. Dice and IoU summarize overlap after a binary decision, AUROC summarizes probability ranking across thresholds, patch-level balanced accuracy supports DIAGSET-style recognition comparisons, and patient-level aggregation exposes whether a method behaves consistently across cases rather than only across pooled pixels [26, 9, 27, 28]. This distinction is central to the present paper because false positives in negative patients remain scientifically meaningful even when aggregate segmentation metrics are high.

### 3. Materials and Methods

#### 3.1. Workflow Overview

The workflow is organized as ten numbered stages connected by explicit files, database rows, and metadata sidecars. Figure 1 summarizes the main method path from patch extraction and preprocessing to training, on-the-fly processing, validation-only optimization, and frozen inference. Early stages build the dataset contract; middle stages curate, split, and sample the data; late stages train single models, optimize a validation-only ensemble recipe, and evaluate that recipe on held-out test patients. Root scripts orchestrate each

stage, while reusable logic lives in domain-specific helper modules.

#### 3.2. Dataset and Curation

DIAGSET was used as the experimental dataset [14]. The ingestion stage registered 429 .ndpi images and 429 .ndpa annotations, of which 419 were paired and validated through converted GeoJSON annotations. Ten images were rejected or failed before final extraction. Patches were extracted with a 224-pixel window, 112-pixel stride, annotation-overlap threshold of 1.0, and minimum tissue-content threshold of 0.3.

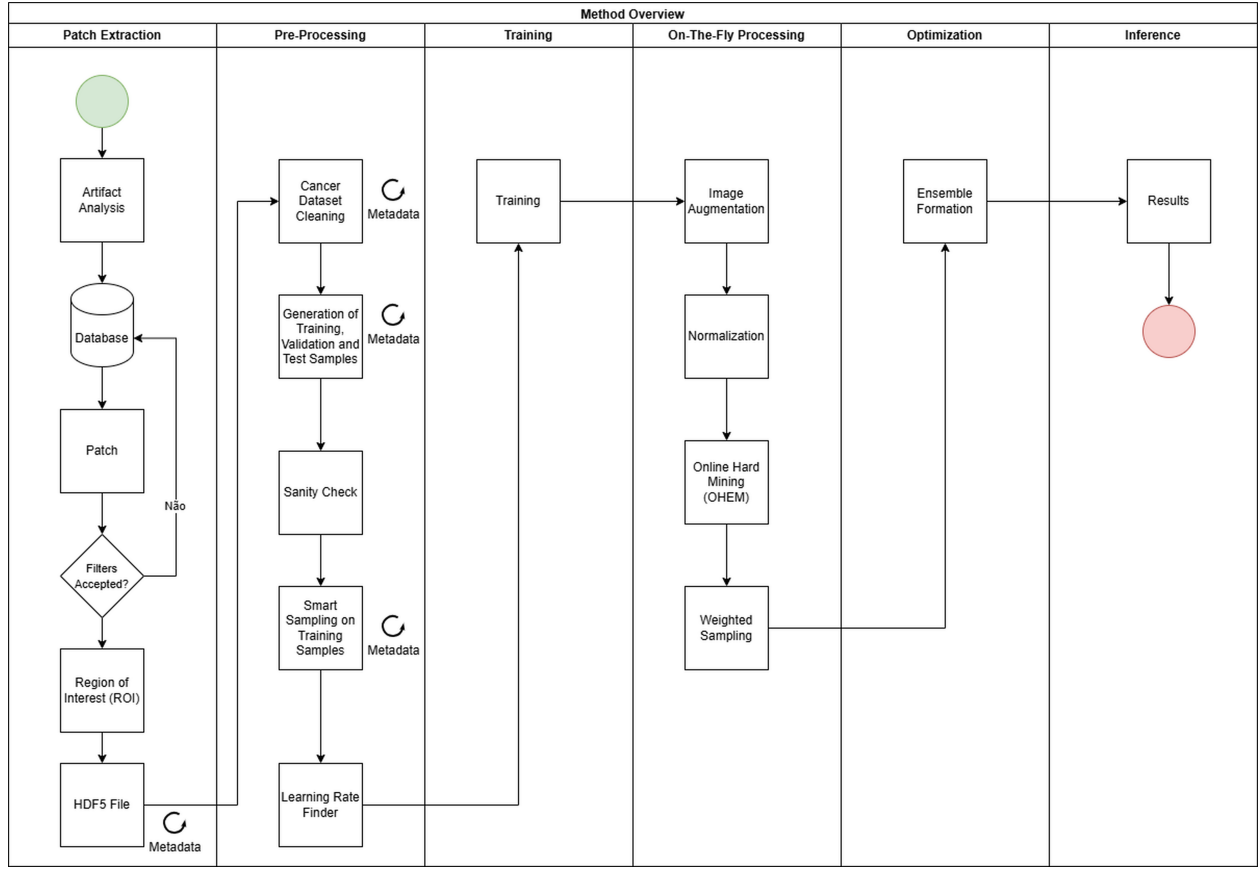
Cancer-patch curation used a small manually reviewed pilot set and a larger labeled master set to tune a graph-based background-contamination rule. The pilot set contained 100 reviewed samples, with 94 approved and 6 rejected. The master set contained 385 samples, with 367 approved and 18 rejected. The selected graph-cleaning configuration used a background-intensity threshold of 167, Felzenszwalb scale parameter  $k = 140$ , minimum segment size 97, a 3-pixel erosion radius, and contamination threshold  $\tau = 0.19$  [10]. Applying this rule rejected 74,047 cancer patches and retained 1,186,778 patches for downstream analysis. Non-cancer patches were not rejected by this graph-cleaning step because the rule was targeted at positive regions contaminated by background or annotation mismatch.

This graph-cleaning step is treated as a pre-split annotation and background-curation operation, not as a downstream segmentation model selected from test performance. The current manuscript evidence records the reviewed sample counts and selected graph parameters, but the preserved result bundle does not provide a complete reviewed-stem-to-final-split audit. Consequently, the cleaning step is not claimed to have been tuned only on final training patients; it is reported as a pre-split curation choice and interpreted as a limitation of the no-rerun evidence package.

#### 3.3. Patient-Level Splitting and Runtime Normalization

After cleaning, the retained DIAGSET patches were split at the stored patient-identifier level. The final dataset contained 393 patients: 236 for training, 78 for validation, and 79 for testing. The test partition contained 387,325 patches from 38 positive and 41 negative patients. This split policy is intended to prevent patches associated with the same stored patient identifier from appearing in more than one partition, while preserving the limitation that the repository does not independently prove biological patient identity beyond the identifiers available at ingestion.

Stain normalization is selected downstream through `RUNTIME_NORMALIZATION_METHOD` rather than being baked into separate split assignments. Stage 4 constructs training-derived template and runtime-normalization state for Reinhard, Ruifrok, Macenko, and Vahadane, while the non-normalized condition is treated as a no-transform setting. This design allows all normalization strategies to reuse the same patient split and avoids deriving preprocessing state



**Figure 1:** Overall method used in the DIAGSET experiment. The workflow connects patch extraction, artifact analysis, HDF5-backed storage, cancer-patch cleaning, patient-level train/validation/test generation, sanity checking, train-only smart sampling, learning-rate selection, supervised training, runtime image augmentation and normalization, weighted sampling, validation-only ensemble formation, and frozen inference.

from validation or test patients. The Vahadane experiment used `RUNTIME_VAHADANE_BACKEND=fixed_source`, a fixed-source approximation to per-patch Vahadane normalization; the reported Vahadane results should therefore be interpreted under that operational contract.

### 3.4. Training and Checkpoint Selection

Smart sampling was applied only to the training split. Of 660,867 available training patches, 180,381 were retained, corresponding to 27.29% of the training rows. Validation and test rows were not compacted or filtered, so sampling changed the effective training set without changing final evaluation data. The sampling stage used Phikon-v2 embeddings and stored selected training rows in compact `TRAIN_SELECTED_COMPACT_DIR` storage for faster downstream reads [3, 19, 12].

Each normalization setting was evaluated with the same eight registered architecture–encoder pairs, producing 40 final model trainings. All final trainings used seed 24. The model registry fixed the architecture search space, while training used a weighted binary-cross-entropy plus Dice objective derived from the implemented loss family, inverse-frequency patch sampling, and validation-driven checkpoint selection [13, 12]. The loss in Eq. (1) was

$$\mathcal{L} = \alpha \mathcal{L}_{\text{BCE,fg}} + \beta \mathcal{L}_{\text{Dice,bg}} + \gamma \mathcal{L}_{\text{Dice,fg}}, \quad (1)$$

with  $\alpha = 0.125$ ,  $\beta = 0.157$ , and  $\gamma = 0.173$  in the final protocol. Checkpoints were selected by validation pixel-level AUPRC, with AUROC and MCC tracked as auxiliary metrics [26, 16].

### 3.5. Two-Stream Ensemble Inference

For each normalization setting, the final inference recipe combined eight trained models into a `two_stream_spatial_gating` ensemble. Four models formed a semantic stream used to define a coarse region of interest, and four models formed a spatial stream used for finer segmentation inside that semantic support. The semantic stream optimized normalized model weights and an ROI threshold on validation patients, favoring positive-patient ROI recall while discouraging empty, overly permissive, and negative-patient ROIs. The spatial stream then optimized model weights under the fixed ROI by maximizing positive-patient AUPRC inside the ROI while penalizing probability spill outside the ROI and prediction mass on negative validation patients [1, 2, 26].

Let  $P_{\text{sem}}(x)$  and  $P_{\text{spa}}(x)$  be the semantic- and spatial-stream foreground probabilities at pixel  $x$ . The semantic ROI in Eq. (2) is generated by thresholding the downsampled semantic prediction and upsampling it back to patch resolution,

**Table 1**

Validation optimization burden recorded by the no-rerun optimizer artifacts. Each normalization used the same 78 validation patients; 25% of them, corresponding to 20 patients, were used for final calibration. Semantic and spatial columns report the number of optimization trials for each stream. The calibration outcome translates the frozen optimizer status into reviewer-facing language; the exact machine-readable codes are preserved in the evidence index.

| Normalization | Validation patients | Calibration patients | Semantic trials | Spatial trials | Calibration outcome                                  |
|---------------|---------------------|----------------------|-----------------|----------------|------------------------------------------------------|
| None          | 78                  | 20                   | 50              | 50             | Required micro-Dice guardrail could not be satisfied |
| Reinhard      | 78                  | 20                   | 50              | 50             | Calibration target met                               |
| Ruifrok       | 78                  | 20                   | 50              | 50             | Calibration target met                               |
| Macenko       | 78                  | 20                   | 50              | 50             | Required micro-Dice guardrail could not be satisfied |
| Vahadane      | 78                  | 20                   | 50              | 50             | Calibration target could not be satisfied            |

$$R(x) = \text{Up}\left(\mathbf{1}_{\{\text{Down}(P_{\text{sem};s}) > \tau_{\text{roi}}\}}\right)(x). \quad (2)$$

The final gated prediction in Eq. (3) is then

$$P_{\text{final}}(x) = P_{\text{spa}}(x)R(x), \quad \hat{Y}(x) = \mathbf{1}_{\{P_{\text{final}}(x) > \tau_{\text{dec}}\}}. \quad (3)$$

The final binary decision was calibrated on validation patients with a balanced calibration search rather than a single threshold-only criterion. Each candidate jointly specified the decision threshold, a minimum connected-component area in pixels, a minimum connected-component area as a fraction of patch area, a minimum number of predicted-positive patches per patient, and a minimum predicted-positive area fraction per patient. Before candidate scoring, per-patch connected components below the effective area threshold were removed and the resulting counts were accumulated at the patient level. The calibration objective in Eq. (4) combined Dice on positive calibration patients with a clean-negative indicator on negative calibration patients,

$$\text{Score}_{\text{cal}} = \frac{\sum_{j \in \mathcal{P}_+^{\text{cal}}} \text{Dice}_j + \sum_{j \in \mathcal{P}_-^{\text{cal}}} \mathbf{1}_{\{FP_j=0\}}}{|\mathcal{P}_+^{\text{cal}}| + |\mathcal{P}_-^{\text{cal}}|}. \quad (4)$$

Candidate selection also enforced baseline-relative positive-patient Dice and true-positive-rate guardrails, minimum micro Dice, minimum macro precision, minimum macro true-positive rate, and a target negative-clean rate. When these targets were infeasible, the recipe recorded the calibration status rather than silently treating the selected operating point as satisfying all constraints. Table 1 summarizes the validation optimization burden recorded in the frozen optimizer artifacts. The frozen ROI thresholds were 0.2719, 0.2808, 0.2104, 0.1748, and 0.2268, and the final decision thresholds were 0.5900, 0.5000, 0.6000, 0.5700, and 0.5500 for no normalization, Reinhard, Ruifrok, Macenko, and Vahadane, respectively. These thresholds summarize only part of the frozen recipe; the recipe also stores the validation-selected component, patch-count, patient-area post-processing parameters, and negative false-positive area diagnostics.

### 3.6. Metrics and Uncertainty

Final evaluation was performed only on the frozen test split. Pixel-level counts were accumulated after the validation-selected thresholding, connected-component filtering, and frozen patient-level suppression rules. By contrast, AUROC was computed from raw foreground probabilities before hard post-processing, so AUROC and post-processed Dice/NCR evaluate related but not identical prediction objects. Micro metrics were computed by pooling pixel counts across test patients before evaluating the metric, whereas macro metrics were computed per patient and then averaged. For macro Dice, a negative patient with neither annotated lesion pixels nor predicted positive pixels contributed Dice equal to 1, whereas a negative patient with any false-positive pixels contributed Dice equal to 0. The reported summary therefore separates positive-patient Dice from negative-clean rate to avoid hiding empty-mask behavior inside one macro overlap value. Positive-patient Dice was computed only among patients with annotated lesion pixels. Negative-clean rate, defined in Eq. (5), was the proportion of negative patients with no positive prediction,

$$\text{NCR} = \frac{1}{|\mathcal{P}_-|} \sum_{j \in \mathcal{P}_-} \mathbf{1}_{\{FP_j=0\}}, \quad (5)$$

where  $\mathcal{P}_-$  is the set of negative patients and  $FP_j$  is the number of false-positive pixels for patient  $j$ .

Stage 10 also emits a DIAGSET-style patch-recognition summary without changing the segmentation decision path. For patch  $i$ , the prediction is converted to a binary patch label from the component-filtered segmentation mask using Eq. (6),

$$\hat{z}_i = \mathbf{1}_{\{a(\hat{Y}_i) > \tau_{\text{patch}}\}}, \quad (6)$$

where  $a(\hat{Y}_i)$  is the predicted positive area fraction and the default  $\tau_{\text{patch}} = 0$  marks a patch as cancer when any positive mask area remains after component filtering. Ground-truth patch labels are taken from the manifest label produced by the extraction-overlap rule, with a binary-mask fallback when labels are absent from the batch. The patch report gives point estimates for Accuracy, Sensitivity, Specificity, and



balanced average accuracy as defined in Eq. (7),

$$\text{AvAcc} = \frac{\text{Sensitivity} + \text{Specificity}}{2}, \quad (7)$$

plus support and confusion counts. The primary DIAGSET-style comparison is recorded before patient-level suppression, and a second section reports the same patch metrics after the frozen patient-level suppression rule.

When confidence intervals were available, they were computed by nonparametric bootstrap with 10,000 resamples at the patient level [7]. This patient-wise procedure was used for micro metrics, macro metrics, and the positive/negative-patient decomposition. AUROC is reported as a point estimate only because the final artifacts preserved the information needed for histogram-based AUROC but did not preserve patient-indexed probability distributions required for a patient-level AUROC bootstrap.

## 4. Results

### 4.1. Curation, Split, and Sampling Outcomes

The preprocessing and curation stages produced a dataset large enough for model development while preserving patient-level evaluation. Starting from 429 registered image–annotation pairs, 419 cases passed pairing and annotation validation. Graph cleaning removed 74,047 positive patches and left 1,186,778 accepted patches. The resulting patient-level split preserved similar positive and negative patient counts: 115 positive and 121 negative patients in training, 43 positive and 35 negative patients in validation, and 38 positive and 41 negative patients in testing. Smart sampling then reduced only the training split from 660,867 to 180,381 patches.

These numbers are methodologically important because they define the evaluation contract. Curation occurred before patient splitting, smart sampling was restricted to training rows, checkpoints were selected on validation data, ensemble weights and thresholds were optimized on validation patients, and the test set remained held out until frozen inference.

### 4.2. Final Test Performance

Table 2 summarizes the final held-out test results for the five normalization strategies. The table is intentionally compact: it reports the metrics needed to interpret the main segmentation and patient-level findings without listing every per-model checkpoint and auxiliary metric. Values should be read as point estimates with available patient-bootstrap intervals, not as formal pairwise superiority tests. Figure 2 visualizes the corresponding micro- and macro-averaged metric profiles, and Table 3 adds the DIAGSET-style patch-recognition summary for comparisons that require patch-level Accuracy or AvAcc rather than segmentation overlap.

The largest point estimate depends on the metric. The non-normalized condition produced the largest micro Dice and macro Dice point estimates, describing the strongest observed overlap both after pooling all test pixels and after

averaging patient-level results. Ruifrok produced the largest AUROC point estimate and the largest Dice among positive patients, describing stronger observed probability ranking and favorable behavior on lesion-bearing cases. Because several intervals overlap and AUROC lacks a patient-level interval in the available artifacts, these observations should be interpreted as metric-specific descriptive findings rather than proof that one normalization is statistically superior.

The patient-level decomposition exposes the main limitation and trade-off. Ruifrok, Macenko, and the non-normalized setting reached positive-patient Dice values near 0.78–0.80, indicating that the workflow could localize annotated lesion regions in many positive cases. However, stronger positive-patient behavior did not always coincide with conservative negative-patient behavior: the non-normalized and Reinhard settings suppressed all positive predictions in 30 of 41 negative patients, whereas Vahadane did so in only 6 of 41. Even the most conservative settings therefore left positive predictions in 11 negative test patients.

### 4.3. Contextual Comparisons

The patch-recognition results can first be placed beside the original DIAGSET binary S1 setting, but only as a protocol-aware reference. The DIAGSET paper used its own extraction protocol, magnification settings, and classifier ensemble, so Table 4 should not be read as a strict reproduction of that experiment [14]. Under this caveat, the largest observed workflow patch Accuracy was 0.9325 with Ruifrok, 1.33 percentage points below the DIAGSET S1 Accuracy of 0.9458, while the largest observed workflow AvAcc was 0.9009 with Macenko, 4.61 percentage points below the DIAGSET S1 AvAcc of 0.9470.

Pathologist readings are not tabulated as numeric model comparators in this manuscript because the workflow row is patch-level recognition before patient-level suppression, whereas published prostate pathology studies evaluate WSI, biopsy, slide, or case-level diagnostic decisions [21, 22, 8]. Those studies remain clinically relevant context, but the present DIAGSET experiment does not provide a commensurate human-versus-model benchmark.

## 5. Discussion

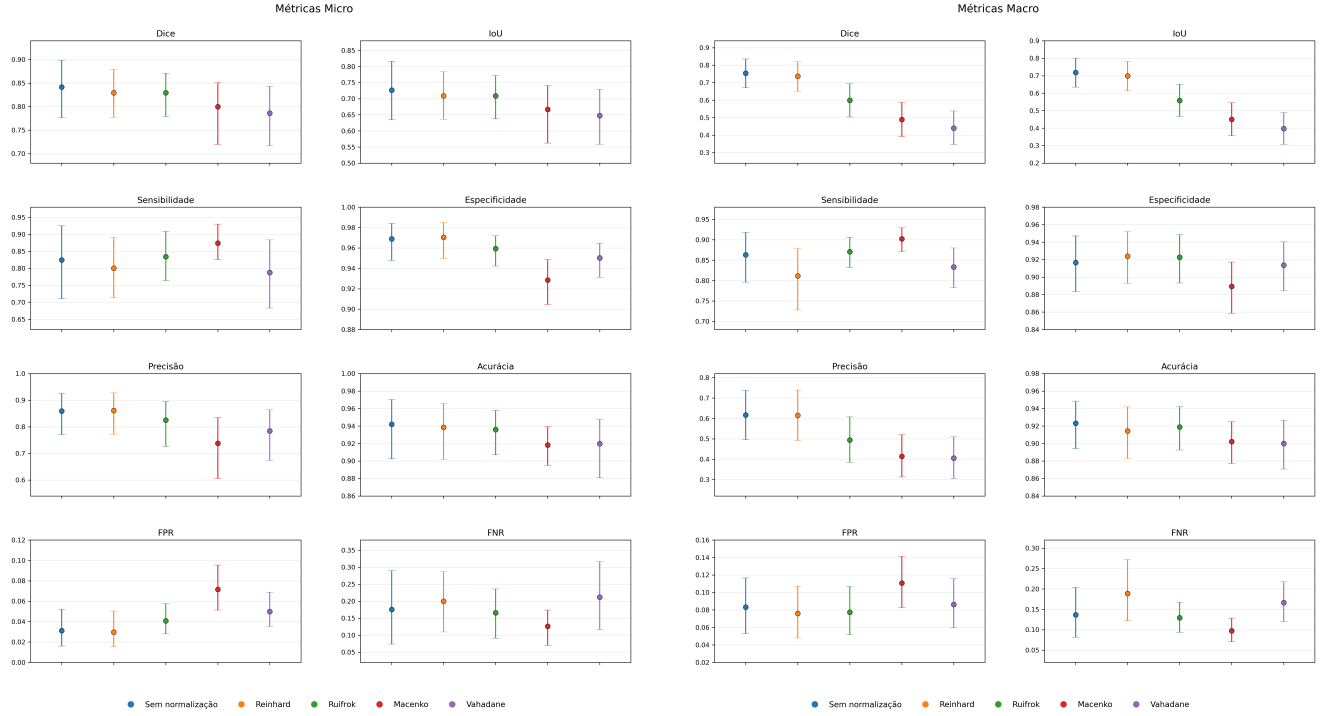
The DIAGSET experiment supports three paper-level conclusions within a single-dataset scope. First, the proposed workflow can compare preprocessing and normalization choices under a traceable, patient-level protocol. The comparison is not a loose collection of model runs: curation, split assignment, smart sampling, checkpoint selection, ensemble composition, threshold calibration, and final inference are connected through explicit artifacts and compatibility checks.

Second, color normalization did not produce a uniform descriptive pattern. No normalization produced the largest point estimates for pixel-pooled and patient-averaged Dice, Ruifrok produced the largest AUROC point estimate and positive-patient Dice, and Macenko produced the largest DIAGSET-style patch AvAcc. This pattern agrees with the

**Table 2**

Final DIAGSET test performance by runtime-normalization strategy. Dice intervals are 95% patient-bootstrap confidence intervals. AUROC is reported as a point estimate because patient-indexed AUROC bootstrap artifacts were not available. NCR denotes negative-clean rate, the fraction of negative patients with no positive prediction.

| Normalization | Micro Dice              | Macro Dice              | AUROC  | Positive-patient Dice   | NCR                     |
|---------------|-------------------------|-------------------------|--------|-------------------------|-------------------------|
| None          | 0.8415 [0.7767, 0.8987] | 0.7540 [0.6710, 0.8356] | 0.9508 | 0.7781 [0.6816, 0.8628] | 0.7317 [0.5909, 0.8649] |
| Reinhard      | 0.8296 [0.7773, 0.8789] | 0.7363 [0.6510, 0.8192] | 0.9445 | 0.7413 [0.6385, 0.8327] | 0.7317 [0.5909, 0.8649] |
| Ruifrok       | 0.8295 [0.7786, 0.8710] | 0.5993 [0.5038, 0.6948] | 0.9579 | 0.7985 [0.7189, 0.8688] | 0.4146 [0.2666, 0.5641] |
| Macenko       | 0.8000 [0.7198, 0.8510] | 0.4894 [0.3921, 0.5878] | 0.9534 | 0.7806 [0.6930, 0.8585] | 0.2195 [0.1000, 0.3500] |
| Vahadane      | 0.7862 [0.7167, 0.8433] | 0.4405 [0.3454, 0.5378] | 0.9389 | 0.7579 [0.6716, 0.8362] | 0.1463 [0.0476, 0.2632] |



**Figure 2:** Micro- and macro-averaged DIAGSET test metrics across normalization strategies. Micro metrics pool pixel counts before metric computation, whereas macro metrics compute patient-level values before averaging. Error bars denote 95% patient-bootstrap confidence intervals where available.

broader caution that stain preprocessing choices can change model behavior without guaranteeing a single dominant strategy across all metrics [29]. In this experiment, the choice of metric changes which configuration appears most favorable, but the available artifacts do not support a formal claim of statistical superiority among normalization methods.

Third, patient-level false positives remain the primary limitation. High Dice in positive patients is not sufficient for a clinically conservative interpretation if negative patients often receive some positive prediction. The current results therefore support methodological viability and reproducible

**Table 3**

DIAGSET-style patch recognition on the held-out test set. The primary comparison is before patient-level suppression; the last column reports AvAcc after applying the frozen patient-level suppression rule.

| Normalization | Accuracy | AvAcc  | Sensitivity | Specificity | Post-supp. AvAcc |
|---------------|----------|--------|-------------|-------------|------------------|
| None          | 0.9323   | 0.8923 | 0.8283      | 0.9562      | 0.8970           |
| Reinhard      | 0.9296   | 0.8811 | 0.8036      | 0.9586      | 0.8854           |
| Ruifrok       | 0.9325   | 0.8997 | 0.8474      | 0.9520      | 0.9005           |
| Macenko       | 0.9166   | 0.9009 | 0.8758      | 0.9260      | 0.9015           |
| Vahadane      | 0.9183   | 0.8715 | 0.7968      | 0.9462      | 0.8715           |

**Table 4**

Contextual comparison between the workflow patch-recognition results and the original DIAGSET binary S1 setting. The comparison is not direct because the original paper used different extraction, magnification, and classifier choices.

| Metric   | DIAGSET S1 | Largest workflow value | Gap        | Interpretation                                               |
|----------|------------|------------------------|------------|--------------------------------------------------------------|
| Accuracy | 0.9458     | 0.9325, Ruifrok        | -1.33 p.p. | Close in aggregate patch accuracy under a different protocol |
| AvAcc    | 0.9470     | 0.9009, Macenko        | -4.61 p.p. | Larger balance gap between sensitivity and specificity       |

comparison, but not standalone clinical deployment. Improving calibration, patient-level post-processing, and false-positive suppression should be prioritized before extending claims beyond this experimental setting.

The study also has clear scope limits. DIAGSET is a single prostate histopathology dataset, and the workflow has not yet been validated here on other organs, scanner distributions, annotation protocols, or cancer types. The stored patient identifiers enforce split isolation within the repository contract, but they do not independently verify biological patient uniqueness outside the source metadata. The validation set carried substantial optimization burden because checkpoints, ensemble composition, ROI thresholds, decision thresholds, and patient-level post-processing were selected before frozen test inference. In addition, the available no-rerun artifacts evaluate the integrated workflow and normalization settings, not independent causal contributions of graph cleaning, smart sampling, two-stream gating, or patient suppression. Finally, the Vahadane condition corresponds to the fixed-source runtime approximation used in the experiment, not exact per-patch Vahadane fitting.

## 6. Conclusion

This paper presented a reproducibility-oriented workflow for whole-slide prostate cancer segmentation and evaluated it on DIAGSET. The main contribution is methodological: the workflow treats the full experimental chain as the object of study, from slide ingestion and patch curation through patient-level splitting, training, validation-only ensemble optimization, and frozen test inference. On DIAGSET, the workflow produced interpretable metric-specific findings rather than a single universally superior normalization strategy. No normalization produced the largest pixel-pooled and patient-averaged Dice point estimates, Ruifrok produced the largest AUROC point estimate and positive-patient Dice, and Macenko produced the largest DIAGSET-style patch AvAcc. At the same time, persistent false positives in negative patients limit clinical interpretation. Future work should evaluate the same artifact-preserving protocol on external datasets, add repeated-seed or paired-difference analyses when computationally feasible, and develop stronger patient-level calibration and false-positive-control mechanisms.

## CRedit authorship contribution statement

**Bruno Henrique Nunes:** Conceptualization, Data curation, Formal analysis, Methodology, Software, Validation, Visualization, Writing - original draft. **Ana Estela Antunes da Silva:** Conceptualization, Project administration, Supervision, Writing - review and editing.

## Funding

This study was financed in part by the Coordenacao de Aperfeicoamento de Pessoal de Nivel Superior – Brasil (CAPES) – Finance Code 001. The funder had no role in study design, data collection, data analysis, interpretation, writing of the manuscript, or the decision to submit the article for publication.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Ethics Approval and Consent to Participate

This study used DIAGSET, a public de-identified histopathology dataset cited in this article. No new human participants were recruited, no new biological specimens were collected, and the authors did not have access to directly identifying participant information. The analysis was secondary use of public data, participant privacy was preserved, and the work was not a clinical trial.

## Data and Code Availability

The DIAGSET data are publicly available through the dataset publication cited in this article. The workflow code is available at [https://github.com/bhnunes/Master\\_Coding](https://github.com/bhnunes/Master_Coding). Large generated artifacts, including HDF5 datasets, checkpoints, and inference outputs, are not included in the manuscript source package because of file size. The manuscript source package includes a no-rerun evidence index that maps the reported tables to final metric JSON files, optimizer recipe/configuration files, model registry snapshots, and the repository commit used for this revision.



## Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During preparation of this manuscript, the authors used OpenAI Codex/ChatGPT to assist with submission-compliance checking and editorial revision. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the submitted manuscript.

## References

- [1] Akiba, T., Sano, S., Yanase, T., Ohta, T., Koyama, M., 2019. Optuna: A next-generation hyperparameter optimization framework, in: Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, pp. 2623–2631. URL: <https://doi.org/10.1145/3292500.3330701>, doi:10.1145/3292500.3330701.
- [2] Bergstra, J.S., Bardenet, R., Bengio, Y., Kégl, B., 2011. Algorithms for hyper-parameter optimization, in: Advances in Neural Information Processing Systems, pp. 2546–2554. URL: <https://papers.nips.cc/paper/4443-algorithms-for-hyper-parameter-optimization>, accessed 3 June 2026.
- [3] Campanella, G., Chen, S., Singh, M., et al., 2025. A clinical benchmark of public self-supervised pathology foundation models. Nature Communications 16, 3640. URL: <https://doi.org/10.1038/s41467-025-58796-1>, doi:10.1038/s41467-025-58796-1.
- [4] Caruana, R., Niculescu-Mizil, A., Crew, G., Ksikes, A., 2004. Ensemble selection from libraries of models, in: Proceedings of the Twenty-First International Conference on Machine Learning, Association for Computing Machinery, p. 18. URL: <https://doi.org/10.1145/1015330.1015432>, doi:10.1145/1015330.1015432.
- [5] Dawood, M., Branson, K., Tejpar, S., Rajpoot, N., Minhas, F.u.A.A., 2026. Confounding factors and biases abound when predicting molecular biomarkers from histological images. Nature Biomedical Engineering URL: <https://doi.org/10.1038/s41551-026-01616-8>, doi:10.1038/s41551-026-01616-8, advance online publication.
- [6] Dietterich, T.G., 2000. Ensemble methods in machine learning, in: Multiple Classifier Systems, Springer, pp. 1–15. URL: [https://doi.org/10.1007/3-540-45014-9\\_1](https://doi.org/10.1007/3-540-45014-9_1), doi:10.1007/3-540-45014-9\_1.
- [7] Efron, B., 1979. Bootstrap methods: Another look at the jackknife. The Annals of Statistics 7, 1–26. URL: <https://doi.org/10.1214/aos/1176344552>, doi:10.1214/aos/1176344552.
- [8] Eloy, C., Marques, A., Pinto, J., Pinheiro, J., Campelos, S., Curado, M., Vale, J., Polónia, A., 2023. Artificial intelligence-assisted cancer diagnosis improves the efficiency of pathologists in prostatic biopsies. Virchows Archiv 482, 595–604. URL: <https://doi.org/10.1007/s00428-023-03518-5>, doi:10.1007/s00428-023-03518-5.
- [9] Fawcett, T., 2006. An introduction to ROC analysis. Pattern Recognition Letters 27, 861–874. URL: <https://doi.org/10.1016/j.patrec.2005.10.010>, doi:10.1016/j.patrec.2005.10.010.
- [10] Felzenszwalb, P.F., Huttenlocher, D.P., 2004. Efficient graph-based image segmentation. International Journal of Computer Vision 59, 167–181. URL: <https://doi.org/10.1023/B:VISI.0000022288.19776.77>, doi:10.1023/B:VISI.0000022288.19776.77.
- [11] Folk, M., Heber, G., Koziol, Q., Pourmal, E., Robinson, D., 2011. An overview of the HDF5 technology suite and its applications, in: Proceedings of the 2011 EDBT/ICDT Workshop on Array Databases, Association for Computing Machinery, pp. 36–47. URL: <https://doi.org/10.1145/1966895.1966900>, doi:10.1145/1966895.1966900.
- [12] He, H., Garcia, E.A., 2009. Learning from imbalanced data. IEEE Transactions on Knowledge and Data Engineering 21, 1263–1284. URL: <https://doi.org/10.1109/TKDE.2008.239>, doi:10.1109/TKDE.2008.239.
- [13] Khened, M., Kori, A., Rajkumar, H., Krishnamurthi, G., Srinivasan, B., 2021. A generalized deep learning framework for whole-slide image segmentation and analysis. Scientific Reports 11, 11579. URL: <https://doi.org/10.1038/s41598-021-90444-8>, doi:10.1038/s41598-021-90444-8.
- [14] Koziarski, M., Cyganek, B., et al., 2024. DiagSet: A dataset for prostate cancer histopathological image classification. Scientific Reports 14, 2251. URL: <https://doi.org/10.1038/s41598-024-52183-4>, doi:10.1038/s41598-024-52183-4.
- [15] Macenko, M., Niethammer, M., Marron, J.S., Borland, D., Woosley, J.T., Guan, X., Schmitt, C., Thomas, N.E., 2009. A method for normalizing histology slides for quantitative analysis, in: 2009 IEEE International Symposium on Biomedical Imaging: From Nano to Macro, pp. 1107–1110. URL: <https://doi.org/10.1109/ISBI.2009.5193250>, doi:10.1109/ISBI.2009.5193250.
- [16] Matthews, B.W., 1975. Comparison of the predicted and observed secondary structure of T4 phage lysozyme. Biochimica et Biophysica Acta (BBA) - Protein Structure 405, 442–451. URL: [https://doi.org/10.1016/0005-2795\(75\)90109-9](https://doi.org/10.1016/0005-2795(75)90109-9), doi:10.1016/0005-2795(75)90109-9.
- [17] Moshkov, N., Mathe, B., Kertesz-Farkas, A., Hollandi, R., Horvath, P., 2020. Test-time augmentation for deep learning-based cell segmentation on microscopy images. Scientific Reports 10, 5068. URL: <https://doi.org/10.1038/s41598-020-61808-3>, doi:10.1038/s41598-020-61808-3.
- [18] Muñoz-Aguirre, M., Ntasis, V.F., Rojas, S., Guigó, R., 2020. PyHIST: A histological image segmentation tool. PLOS Computational Biology 16, e1008349. URL: <https://doi.org/10.1371/journal.pcbi.1008349>, doi:10.1371/journal.pcbi.1008349.
- [19] Oquab, M., Darcet, T., Moutakanni, T., Vo, H.V., Szafraniec, M., Khalidov, V., Fernandez, P., et al., 2024. DINOv2: Learning robust visual features without supervision. Transactions on Machine Learning Research URL: <https://openreview.net/forum?id=a68Sut6zFt>, accessed 3 June 2026.
- [20] Piccolo, S.R., Frampton, M.B., 2016. Tools and techniques for computational reproducibility. GigaScience 5, 30. URL: <https://doi.org/10.1186/s13742-016-0135-4>, doi:10.1186/s13742-016-0135-4.
- [21] Raciti, P., Sue, J., Ceballos, R., Godrich, R., Kunz, J.D., Kapur, S., Reuter, V., Grady, L., Kanan, C., Klimstra, D.S., Fuchs, T.J., 2020. Novel artificial intelligence system increases the detection of prostate cancer in whole slide images of core needle biopsies. Modern Pathology 33, 2058–2066. URL: <https://doi.org/10.1038/s41379-020-0551-y>, doi:10.1038/s41379-020-0551-y.
- [22] Raciti, P., Sue, J., Retamero, J.A., Ceballos, R., Godrich, R., Kunz, J.D., Casson, A., Thiagarajan, D., Ebrahimzadeh, Z., Viret, J., Lee, D., Schüffler, P.J., DeMuth, G., Gulturk, E., Kanan, C., Rothrock, B., Reis-Filho, J., Klimstra, D.S., Reuter, V., Fuchs, T.J., 2023. Clinical validation of artificial intelligence-augmented pathology diagnosis demonstrates significant gains in diagnostic accuracy in prostate cancer detection. Archives of Pathology & Laboratory Medicine 147, 1178–1185. URL: <https://doi.org/10.5858/ARPA.2022-0066-OA>, doi:10.5858/ARPA.2022-0066-OA.
- [23] Reinhard, E., Ashikhmin, M., Gooch, B., Shirley, P., 2001. Color transfer between images. IEEE Computer Graphics and Applications 21, 34–41. URL: <https://doi.org/10.1109/38.946629>, doi:10.1109/38.946629.
- [24] Roberts, D.R., Bahn, V., Ciuti, S., Boyce, M.S., Elith, J., et al., 2017. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. Ecography 40, 913–929. URL: <https://doi.org/10.1111/ecog.02881>, doi:10.1111/ecog.02881.
- [25] Ruifrok, A.C., Johnston, D.A., 2001. Quantification of histochemical staining by color deconvolution. Analytical and Quantitative Cytology and Histology 23, 291–299. URL: <https://pubmed.ncbi.nlm.nih.gov/11815294/>.
- [26] Saito, T., Rehmsmeier, M., 2015. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. PLOS ONE 10, e0118432. URL: <https://doi.org/10.1371/journal.pone.0118432>, doi:10.1371/journal.pone.0118432.
- [27] Seghier, M.L., 2024. Image segmentation evaluation with the dice index: Methodological issues. International Journal of Imaging Systems and Technology 34, e23203. URL: <https://doi.org/10.1002/ima.23203>, doi:10.1002/ima.23203.

- [28] Sokolova, M., Lapalme, G., 2009. A systematic analysis of performance measures for classification tasks. *Information Processing & Management* 45, 427–437. URL: <https://doi.org/10.1016/j.ipm.2009.03.002>, doi:10.1016/j.ipm.2009.03.002.
- [29] Tellez, D., Litjens, G., Bándi, P., Bulten, W., Bokhorst, J.M., Ciompi, F., van der Laak, J., 2019. Quantifying the effects of data augmentation and stain color normalization in convolutional neural networks for computational pathology. *Medical Image Analysis* 58, 101544. URL: <https://doi.org/10.1016/j.media.2019.101544>, doi:10.1016/j.media.2019.101544.
- [30] Vahadane, A., Peng, T., Sethi, A., Albarqouni, S., Wang, L., Baust, M., Steiger, K., Schlitter, A.M., Esposito, I., Navab, N., 2016. Structure-preserving color normalization and sparse stain separation for histological images. *IEEE Transactions on Medical Imaging* 35, 1962–1971. URL: <https://doi.org/10.1109/TMI.2016.2529665>, doi:10.1109/TMI.2016.2529665.
- [31] Weng, Z., Seper, A., Pryalukhin, A., Mairinger, F., Wickenhauser, C., Bauer, M., Glamann, L., et al., 2024. GrandQC: A comprehensive solution to quality control problem in digital pathology. *Nature Communications* 15, 10685. URL: <https://doi.org/10.1038/s41467-024-54769-y>, doi:10.1038/s41467-024-54769-y.
- [32] Wilkinson, M.D., Dumontier, M., Aalbersberg, I.J., Appleton, G., Axton, M., et al., 2016. The FAIR guiding principles for scientific data management and stewardship. *Scientific Data* 3, 160018. URL: <https://doi.org/10.1038/sdata.2016.18>, doi:10.1038/sdata.2016.18.